

UNIT 4 – STATISTICAL INFERENCE

CO4–AT3 – VIVA VOCE QUESTIONS & ANSWERS

Assessment Topic: Statistical Inference, Frequentist Approach, Point Estimation, Variability in Estimates, Confidence Intervals, and Hypothesis Testing

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Course / Unit	Unit 4 – Statistical Inference
Assessment	CO4–AT3 – Viva Voce
Coverage	Questions 1–50
Answer	viva-oriented

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using information from a sample to draw conclusions or make predictions about a larger population. It mainly involves estimation and hypothesis testing.

2. Why is statistical inference important in data science?

It helps data scientists make reliable conclusions from limited data, quantify uncertainty, compare alternatives, and support evidence-based decisions without needing data from the entire population.

3. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize and present the observed data using measures such as mean, median, standard deviation, tables, and graphs. Inferential statistics use sample data to make conclusions or predictions about a population.

4. What is a population in statistics?

A population is the complete set of individuals, observations, objects, or measurements about which a statistical study is concerned.

5. What is a sample?

A sample is a subset of observations selected from a population for analysis. It is used to obtain information about the population.

6. Why do data scientists usually work with samples instead of complete populations?

Studying an entire population may be costly, time-consuming, impractical, or impossible. A representative sample can provide useful information with substantially fewer resources.

7. What is a population parameter?

A population parameter is a numerical characteristic of a population, such as the population mean μ , population proportion p , or population variance σ^2 .

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data, such as the sample mean $\bar{x}$, sample proportion $\hat{p}$, or sample variance s^2 .

9. Give a real-world example where statistical inference is used in industry.

A manufacturing company may inspect a sample of products to estimate the defect rate of the entire production line. The sample results are used to make decisions about overall product quality.

10. What are the two major activities involved in statistical inference?

The two major activities are estimation, which estimates unknown population parameters, and hypothesis testing, which evaluates claims about population parameters.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

The frequentist approach treats population parameters as fixed but unknown quantities and bases inference on the long-run behavior of statistics calculated from repeated random samples.

12. What is the basic idea behind frequentist statistics?

The basic idea is that probability describes the long-run frequency of outcomes from a random process. Evidence about an unknown parameter is obtained from sample data and sampling distributions.

13. How does the frequentist approach differ from the Bayesian approach?

Frequentist inference treats parameters as fixed unknown values and uses repeated-sampling ideas. Bayesian inference represents uncertainty about parameters using probability distributions and combines prior information with observed data.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered fixed but unknown, while the observed data are viewed as one outcome from a random sampling process.

15. What does long-run frequency mean in frequentist statistics?

It refers to the proportion or frequency with which an event would occur if the same random experiment or sampling procedure were repeated a very large number of times.

16. Why is random sampling important in the frequentist approach?

Random sampling helps reduce systematic selection bias and provides a probability-based sampling distribution, allowing uncertainty and error in statistical estimates to be quantified.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In the frequentist framework, the population parameter is treated as a fixed but unknown constant. Randomness is associated with the sample and the statistic calculated from it.

18. Give an example of a frequentist problem in a data science application.

A data scientist may test whether the average response time of a new software version is less than a specified target. A random sample of response times is analyzed using a confidence interval or hypothesis test.

C. Point Estimation

19. What is a point estimate?

A point estimate is a single numerical value calculated from sample data and used as an estimate of an unknown population parameter.

20. What is an estimator?

An estimator is a statistical rule or formula that uses sample data to estimate a population parameter. For example, $\bar{x}$ is an estimator of the population mean μ .

21. What is the difference between an estimator and an estimate?

An estimator is the rule or statistic used for estimation, while an estimate is the numerical value obtained after applying that estimator to a particular sample.

22. What is the point estimator for the population mean?

The sample mean, $\bar{x} = (\sum x_i)/n$, is the standard point estimator for the population mean μ .

23. How do you estimate a population proportion from a sample?

The population proportion p is estimated using the sample proportion $\hat{p} = x/n$, where x is the number of observations with the characteristic of interest and n is the sample size.

24. What is the point estimate of population variance?

The usual unbiased point estimator of population variance σ^2 is the sample variance, $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$.

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean uses all observations, is unbiased for the population mean under random sampling, and generally becomes more precise as the sample size increases.

26. What properties should a good estimator have?

Important properties include unbiasedness, consistency, efficiency, and reliability. Ideally, the estimator should have small bias and low variance.

27. What is an unbiased estimator?

An estimator is unbiased if its expected value equals the true population parameter. For example, $E(\bar{x}) = \mu$, so the sample mean is an unbiased estimator of μ .

28. What is meant by the bias of an estimator?

Bias is the difference between the expected value of an estimator and the true parameter. Mathematically, $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$.

29. What is the difference between bias and variance of an estimator?

Bias measures systematic deviation of an estimator's expected value from the true parameter, while variance measures how much the estimator varies across repeated samples.

30. Why can two different samples produce different point estimates for the same population parameter?

Because samples contain different observations due to random sampling variation. Therefore, statistics such as the sample mean or proportion can differ from one sample to another.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

A point estimate is based on a sample rather than the entire population. Since different random samples can produce different estimates, the estimate is subject to sampling variability and uncertainty.

32. What is the sampling distribution of an estimator?

It is the probability distribution of a statistic obtained from all possible random samples of a specified size from the same population.

33. What is the standard error?

The standard error is the standard deviation of the sampling distribution of an estimator. It measures the typical amount by which an estimator varies from sample to sample.

34. How is standard error different from standard deviation?

Standard deviation measures variability among individual observations in a dataset. Standard error measures variability of a statistic across repeated samples.

35. How does increasing the sample size affect the standard error of the sample mean?

For independent observations, the standard error of the sample mean is approximately $SE(\bar{x}) = \sigma/\sqrt{n}$. Therefore, increasing n decreases the standard error.

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more information about the population and averages out random fluctuations more effectively, making the estimate more stable.

37. What is the relationship between standard error and confidence intervals?

The standard error determines how much the estimate is expected to vary. For a given confidence level, a larger standard error generally produces a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No. Different sample means are expected because of sampling variability. Both samples can be valid if they were collected properly, even though they produce different estimates.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a range of plausible values for an unknown population parameter, calculated from sample data together with a specified confidence level.

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, whereas a confidence interval provides a range and indicates the precision and uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

If the same sampling procedure were repeated many times and a 95% confidence interval were constructed each time, approximately 95% of those intervals would contain the true population parameter.

42. What is the difference between a confidence level and a confidence interval?

The confidence level, such as 95%, describes the long-run performance of the interval procedure. The confidence interval is the specific numerical range calculated from one sample.

43. What factors determine the width of a confidence interval?

The main factors are sample size, variability in the data, and the selected confidence level. Greater variability and higher confidence levels generally increase width, while larger sample sizes decrease width.

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size generally decreases the standard error, so the confidence interval becomes narrower and the estimate becomes more precise.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

The interval becomes wider because a higher confidence level requires a larger critical value, providing greater long-run coverage of the true parameter.

46. What is the difference between a z-confidence interval and a t-confidence interval?

A z-confidence interval uses a critical value from the standard normal distribution, usually when the population standard deviation is known or normal-approximation conditions are appropriate. A t-confidence interval uses the t-distribution and accounts for estimation of the population standard deviation.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

The t-distribution is commonly used when the population standard deviation is unknown and is estimated using the sample standard deviation, particularly for smaller samples when the population is reasonably normal.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical procedure used to evaluate a claim about a population parameter using sample data and a predefined decision rule.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_a)?

The null hypothesis H_0 represents the default or no-effect claim, while the alternative hypothesis H_a represents the claim or effect that the researcher wants to find evidence for. The data are used to determine whether H_0 should be rejected.

50. How can a confidence interval be used to make a decision about a hypothesis?

A confidence interval can be compared with the value specified by H_0 . For example, for a two-sided 95% confidence interval, if the null value lies outside the interval, H_0 is rejected at the 5% significance level. If the null value lies inside the interval, there is insufficient evidence to reject H_0 .

Viva Voce Quick Revision Points

- Statistical inference uses sample information to learn about a population.
- Frequentist inference treats parameters as fixed and uses repeated-sampling behavior.
- A point estimate is a single value; a confidence interval gives a range of plausible values.
- Standard error measures the sampling variability of an estimator.
- Larger samples generally reduce standard error and produce narrower confidence intervals.
- Higher confidence levels produce wider confidence intervals.
- Hypothesis testing evaluates evidence against a null hypothesis.
- A confidence interval can support a hypothesis-testing decision by checking whether the null value lies inside or outside the interval.